



# Radical scavenging gelatin methacrylamide based bioink formulation for three dimensional bioprinting of parenchymal liver construct

Roopesh R. Pai<sup>a,1</sup>, Shilpa Ajit<sup>a,1</sup>, Anupama Sekar J<sup>a</sup>, Sarath S. Nair<sup>b</sup>, Anil Kumar P.R.<sup>a,\*</sup>, Shiny Velayudhan<sup>c,\*\*</sup>

<sup>a</sup> Division of Tissue Culture, India

<sup>b</sup> Division of Extra Corporeal Devices, India

<sup>c</sup> Division of Dental Products, India

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## ABSTRACT

Methacrylated gelatin (GelMA) in the form of methacryloyl, methacrylate, and methacrylamide is an established and widely accepted photocrosslinkable bioink, for three dimensional bioprinting of various tissues. One of the limitations of photocrosslinkable bioinks is the inability to control the free radicals generated by photoinitiators and ultraviolet (UV) rays. The presence of excess free radicals compromises the viability and functionality of cells during crosslinking. In this study, ascorbic acid, a known free radical scavenger (FRS) molecule, was introduced into the GelMA bioink formulation to protect the cell viability, proliferation, and tissue functions of 3D bioprinted parenchymal liver constructs. The concentration of FRS in the bioink was optimized and used for 3D bioprinting of HepG2 cells. The results confirmed that the inclusion of 3.4 mM FRS in the GelMA bioink formulation nullified the excess ROS formed inside the cells. Furthermore, the optimized GelMA formulation containing FRS preserved and improved the cell activity, albumin, and urea synthesis in the 3D construct over 7 days in culture. In the future, this concept could be implemented in the biofabrication of large liver constructs that require multiple or longer durations of UV irradiation.

## 1. Introduction

Three-dimensional (3D) bio-printing has evolved as a promising tool in tissue engineering and regenerative medicine and is considered an advanced technology that helps in engineering structurally similar biological substitutes by precisely positioning the cells in predefined locations. This technology belongs to the family of additive manufacturing techniques that allow rapid prototyping of tissue constructs by layer-by-layer simultaneous deposition of multiple cell types, proteins, and bioactive substances [1,2]. Accurate spatiotemporal control based on a computer-generated design allows a 3D bioprinting strategy to fabricate biological substitutes with organized micro-architecture, interconnected pores, precise distribution of cells inside the structure, and biochemical demands of the tissues and organs. To accomplish this task of cell deposition in accordance with a particular

design, a carrier medium or bioink, that not only supports the cells but also functions as the framework of the bioprinted construct.

A bioink is defined as 'a formulation of cells suitable for processing by an automated biofabrication technology that may also contain biologically active components and biomaterials' [3]. To facilitate enhanced cell function and survivability, bioinks should closely resemble the cell extracellular matrix (ECM) and possess suitable gelling characteristics for different bioprinting approaches are the ideal candidates. In this regard, hydrogels perfectly fit this bill as they are structurally closer to the ECM and can be cross-linked to obtain highly hydrated three-dimensional (3D) networks, are used as excellent scaffolds for tissue engineering scaffolds and are capable of providing mechanical stability to the biofabricated constructs/tissues [4]. Numerous hydrogel systems have been explored as bioinks. However, modified gelatin, known as gelatin methacryloyl, gelatin methacrylamide, and

\* Corresponding author. Department of Applied Biology, Biomedical Technology Wing, Sree Chitra Tirunal Institute for Medical Sciences and Technology Thiruvananthapuram, Kerala, 695 012, India.

\*\* Corresponding author. Department of Biomaterial Science and Technology, Biomedical Technology Wing, Sree Chitra Tirunal Institute for Medical Sciences and Technology Thiruvananthapuram, Kerala, 695 012, India.

E-mail addresses: [anilkumarpr@sctimst.ac.in](mailto:anilkumarpr@sctimst.ac.in) (P.R. Anil Kumar), [shiny@sctimst.ac.in](mailto:shiny@sctimst.ac.in) (S. Velayudhan).

<sup>1</sup> Equal contribution.

gelatin methacrylate, all abbreviated as GelMA is one of the most widely studied hydrogel systems for the bio-fabrication of tissue constructs [5].

GelMA is a photo-crosslinkable polymer that can be readily cross-linked by irradiation at a specific wavelength range in the presence of a photoinitiator (PI), which initiates rapid crosslinking at physiological temperature, pH, and ionic environment and allows for the incorporation of living cells by *in situ* gelation [6]. During photo-crosslinking, PIs generate free radicals upon exposure to a specific wavelength of light and initiate polymerization by cross-linking, which leads to the formation of hydrogels. However, the concentration of PI, inactivated residues, or byproducts formed after cross-linking can elicit toxic effects on cells [7]. The popular photo-initiators used in bioinks are Irgacure (2-hydroxy-1-[4-(2-hydroxyethoxy) phenyl] - 2 -methyl-1-propanone) (Irgacure 2959), Eosin Y, Biokey and VA-086 [8–11]. Among these, Irgacure 2959 elicits minimal cytotoxicity over a wide range of cell types [12] and has been reported safer within a range of 0.2–1 wt percentage for cross-linking GelMA [13].

The free radicals formed during photo-initiated cell encapsulation interact within the cell membrane, proteins, and deoxyribonucleic acids, causing serious cellular damage by affecting many biochemical and physiological activities [14]. One of the adverse effects of free radicals is its influence on the serine/threonine-specific protein kinase known as AKT (v-Akt murine thymoma viral oncogene). Another study by Williams et al. compared UV-sensitive PI on cell types over a range of mammalian species and showed that Irgacure 2959 is well tolerated for photo-encapsulation [12]. However, different cell types displayed variable responses with respect to cell survival and proliferation towards identical concentrations of the same photo-initiator. This is due to the lack of control over excess free radicals generated during photopolymerization. Moreover, UV irradiation generates reactive oxygen species (ROS) within cells that can damage macromolecules such as DNA, proteins, carbohydrates, and lipids. Thus, cells encapsulated in photo-crosslinkable hydrogels encounter exogenous and endogenous cytotoxic radical species. Many enzymatic and non-enzymatic antioxidants are known to eliminate pro-oxidants and scavenge free radicals in tissues [15]. Vital molecules such as tocopherols (vitamin E), ascorbic acid (vitamin C), and carotenoids (vitamin B) are well-known antioxidants in living systems that react with peroxy free radicals and protect biological systems [16]. Hence, we hypothesized that the inclusion of free radical scavenging (FRS) molecules in the hydrogel system will improve cell viability and prevent free radical-induced cell damage.

In this study we report a novel photo-protective bioink formulation for bioprinting functional liver constructs. We have added the ascorbic acid (FRS) as a biologically active component in our bioink. L-ascorbic acid can function as the photo-protective molecule to the bioink formulation, and liver tissue constructs. Human hepatoma cell HepG2 is a commonly used liver cell line which proliferates easily and can secrete liver-specific molecules which are routinely analyzed during liver function test. Therefore, we have chosen HepG2 cells in our study to highlight the potential benefits of GelMA-FRS bioink on bioprinted liver constructs. The role of ascorbic acid as a free radical scavenger is well established, however there are no reports on its use as a biologically active component in any bioink formulations reported so far [17,18]. The bioprinted constructs were evaluated for their functional activity. In this study, we report the development of a photo-protective bioink formulation based on GelMA.

## 2. Materials and methods

### 2.1. Materials

Porcine skin gelatin (Type A, 175 bloom), methacrylic anhydride (MAA), dialysis tubing (12–14 kDa), fluorescein diacetate (FDA), Cell Counting Kit – 8 (CCK-8), collagenase-A, and glutaraldehyde were purchased from Sigma, India. Irgacure 2959 (photo-initiator, 2 - hydroxy - 4 - (2 - hydroxyl ethoxy) - 2 - methyl propiophenone) was

purchased from Ciba Specialty Chemicals, Tarrytown, NJ. Propidium iodide (PI) was procured from HiMedia, India. Syringes and needles were purchased from BD Biosciences, USA; Minimum Essential Medium (MEM), Penicillin-Streptomycin (Pen Strep - 10,000 U/ml), fluorescein isothiocyanate-tagged phalloidin (FITC - phalloidin), Trypsin - Ethelene Diamine Tetra Acetic acid, and fetal bovine serum (FBS) were purchased from Invitrogen, USA. The mouse subcutaneous fibroblast cell line (L-929) was procured from American Type Culture Collections (ATCC, USA), and human liver carcinoma cells (HepG2) were obtained from the National Center for Cell Sciences, India. The human albumin ELISA kit was procured from Bethyl Laboratories, USA, and the urea assay kit was procured from BioChain, USA. All cell culture grade plastic ware such as Petri dishes, T25 flasks, and multi-well plates were purchased from Eppendorf India Limited. The consumables for 3D bioprinting were procured from regenHU, Switzerland.

### 2.2. Synthesis of gelatin methacrylamide

GelMA was synthesized using a modified buffer method [19]. Briefly, 10 g of gelatin was dissolved in 100 ml alkaline buffer (carbonate bicarbonate, CB, pH > 10) at 50 °C. 167 µL of MAA was sequentially added every 30 min to the gelatin solution at 50 °C for 3h. Following every addition of MAA, the pH of the reaction solution was monitored, and CB buffer was added to maintain a pH > 10. At the end of 3h, the pH of the reaction was adjusted to 7.3. The reaction mixture was then dialyzed against distilled water for 4 days at 40 °C and lyophilized. The lyophilized polymer (GelMA) was stored for -20 °C further use. For biological analysis, GelMA was sterilized by ethylene oxide gas 37 °C and stored at - 20 °C until further use.

### 2.3. <sup>1</sup>H Nuclear magnetic resonance spectroscopy (<sup>1</sup>H NMR)

<sup>1</sup>H NMR analysis was performed to confirm the functionalization of gelatin with MAA. Briefly, clear solutions of the sample were prepared by dissolving 10 mg each of gelatin and GelMA in 1 ml deuterium oxide (D<sub>2</sub>O) at 40 °C. The NMR spectrum was obtained using a Bruker Avance 400 MHz FT-NMR spectrometer (Germany).

### 2.4. Degree of functionalization

The degree of functionalization (DoF) was determined using the ninhydrin method reported by Loessner *et al* [20]. Briefly, to 850 µl of ninhydrin solution (2.5 mg/ml ninhydrin in a mixture of 0.5 M sodium citrate monobasic: glycerol at 1: 2 ratio) and 150 µl GelMA solution (10% wt/v in deionized water) was incubated at 50 °C for 12 min. Linear dilutions of gelatin in deionized water were used as the standard solutions. The solution was cooled to room temperature and the absorbance was measured at 570 nm using a spectrophotometer (Biotek, USA). The absorbance obtained for a gelatin concentration of X% in the standard curve was extrapolated to obtain the concentration of unreacted groups in GelMA. The DoF was calculated as 100-X %, considering that the reduction in absorbance was due to methacrylamide functionalization.

$$\text{Degree of functionalization (DoF)} = 1 - \left( \frac{\text{GelMA}}{\text{Gelatin}} \right) \times 100$$

### 2.5. Preparation of gelatin methacrylamide hydrogel

Gelatin methacrylamide hydrogels were prepared by crosslinking methacrylamide groups by photoinitiated chain polymerization. GelMA (10% w/v) solution was prepared in phosphate-buffered saline (PBS) or serum-free MEM containing 1% Irgacure as a photoinitiator at 60 °C until fully dissolved. A requisite amount of this solution was exposed to 365 nm UV light at an intensity of 8–12 mW/cm<sup>2</sup> to initiate crosslinking. The exposure time was varied from 2–7 min depending on the

thickness of the prepared gels. Photo-protective GelMA was prepared by adding various concentrations of FRS(X) (X representing 1, 1.5, and 2.25 or 3.4 mM) to 10% GelMA in serum-free medium containing 1% Irgacure. The range of FRS concentration was selected based on IC50 analysis as described in sections 1.1 and 2.1 of supplementary data. The GelMA-FRS(X) mixture was crosslinked with UV, as described above. The time taken to form the photoprotective hydrogel was noted.

## 2.6. Selection of free radical scavenger concentration

The effective concentration of FRS, expressing photoprotectivity to HepG2 cells, was evaluated by analyzing cell viability within GelMA and GelMA-FRS in a sandwich culture model. Cells were cultured in MEM supplemented with 10% FBS and 1% penicillin-streptomycin antibiotics in a CO<sub>2</sub> incubator (New Brunswick, USA) set at 37 °C with 5% CO<sub>2</sub> and >90% relative humidity. Cells were harvested by trypsinization and cultured in a sandwich model, as illustrated in Fig. 1. The sandwich method is similar to cell encapsulation, with the additional advantage of placing cells on a single plane that makes the cell-based assays and imaging for analysis efficiency. GelMA-FRS(X) systems with various concentrations of FRS, where X = 0, 1, 1.5, 2.2, or 3.4 mM) were prepared and added to 48 well plate (100 µL/well) and cross-linked with UV light. Approximately  $1 \times 10^5$  HepG2 cells were seeded over the cross-linked GelMA-FRS hydrogel and incubated for 2h for cell adhesion. A second layer of 100 µL GelMA-FRS system was overlaid on cells and cross-linking was repeated to entrap cells in a sandwich fashion. Sufficient cell culture medium was added and cells were incubated for 24 h. Cells sandwiched in GelMA (FRS = 0 mM) were considered as the control.

### 2.6.1. Cell viability

Cell viability was analyzed at the end of 24h by live/dead staining. Cells sandwiched within GelMA-FRS(3.4) were incubated for 5 min in serum-free medium containing 0.01% FDA and 0.005% PI and observed under a fluorescence microscope (Leica DMI 6000, Switzerland).

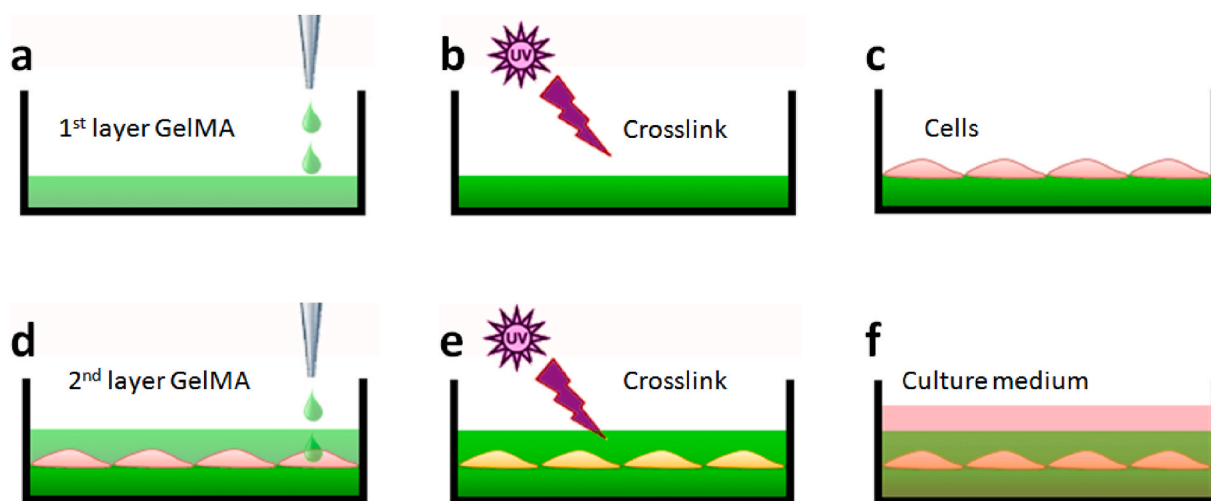
### 2.6.2. Cell proliferation

The proliferation of the cells in the sandwich was estimated from the cell activity expressed in a standard curve plotted with known concentrations of cells. HepG2 cells at different cell densities ( $0.5 \times 10^6$ ,  $1 \times 10^6$ ,  $2 \times 10^6$ ,  $2.5 \times 10^6$  cells/well) were seeded in triplicate into a 24 multi-well plate, incubated for 24h and CCK-8 assay was carried out according to the manufacturer's instructions. Briefly, the culture

medium was removed and the cells were incubated with 500 µL CCK-8 reagent (1:10 dilution in serum-free MEM) for 2h. The medium was then transferred to a clear 96 multi-well plate and the absorbance was measured at 450 nm using a multi-well plate reader (BioTek, USA). Approximately  $1 \times 10^5$  HepG2 cells were sandwiched in the GelMA-FRS (X) system. The cell suspension used for plotting the standard curve was used to estimate the cell number in the sandwich culture within GelMA and GelMA-FRS(X). The cells were cultured for 7 days and proliferation was analyzed on every alternate day (1, 3, 5, and 7 days) using the CCK-8 assay as described above. The number of cells at various concentrations of GelMA-FRS was calculated from the standard curve. The GelMA-FRS(X) system, which supported maximum cell viability and proliferation, was considered for further experiments.

### 2.6.3. Photoprotective effect of gelatin methacrylamide -Free radical scavenger system

Based on the cell proliferation data, GelMA-FRS(3.4) was selected for further evaluation of the photoprotective properties. The reactive oxygen species (ROS) generated within cells suspended in GelMA FRS during UV crosslinking was estimated. The GelMA and GelMA FRS solutions were prepared using HepG2 cells at a concentration of  $5 \times 10^5$  cells/mL and were dispersed in PBS. The solution (200 µL) was transferred to multi-well plates and irradiated with UV light at 365 nm for 3 min to form a hydrogel. Cell monolayers adhered to similar multi-well plates formed from the same cell number were used as controls. Cells exposed to UV were considered as a positive control and the cell monolayer without UV treatment was treated as a negative control. Cells in controls and hydrogels were harvested by incubating with 200 µL of 0.3% collagenase - A in calcium containing HEPES buffer (100 mM HEPES buffer with 20 mM calcium chloride) for 20 min at 37 °C. The cells were transferred to a centrifuge tube and centrifuged (Eppendorf 5430R, India) at 1500 rpm for 2 min to pellet the cells. The supernatant was discarded, and the ROS activities of the cells were determined per product sheet (Lot# AB113851, Abcam). The cell suspension was transferred to a black-bottom-clear 96 well plate and 2', 7' - dichloro-fluorescein diacetate formed due to ROS activity inside the cells was quantified by measuring the fluorescence at excitation/emission of 485 nm/535 nm in a spectrofluorimeter (Synergy H1, Biotek, USA). The relative fluorescence unit (RFU) obtained for the cells from GelMA, GelMA-FRS(3.4), and the positive control was compared with that of the negative control using the following equation:



**Fig. 1.** Illustration showing the steps involved in sandwich culture model: (a) A thin layer of hydrogel was applied first on multiwell plate, (b) Crosslinking of first layer by UV irradiation, (c) Cells are evenly seeded on first layer of hydrogel and allowed to settle for 1 h, (d) A second layer of hydrogel was applied on the cells, (e) Crosslinking of the second layer with UV to entrap cells in sandwich form, (f) Add sufficient culture medium and maintain cells.

$$\text{ROS activity (\%)} = \frac{\text{Test}_{\text{RFU}}}{\text{Negative}_{\text{RFU}}} \times 100$$

#### 2.6.4. Functional evaluation of HepG2 cells in sandwich culture

The effect of the photoprotective nature of GelMA-FRS(3.4) hydrogel in preserving liver-specific functions was determined by analyzing albumin and urea synthesis in HepG2 cells sandwiched within the hydrogel. The HepG2 cells were sandwiched in GelMA-FRS(X) systems and maintained for 7 days with medium changes on days 1, 3, 5, and 7. The spent medium was collected and stored at  $-80^{\circ}\text{C}$  until analysis.

Albumin synthesis by HepG2 cells was analyzed using an ELISA kit (Bethyl Laboratories, USA) according to the manufacturer's instructions. Serial dilutions of standards and the spent culture medium collected at various time points were added in duplicate to the first antibody (100  $\mu\text{L}$ ) coated 96 wells. After 1 h incubation at room temperature, wells were rinsed four times with wash buffer, and the second antibody (100  $\mu\text{L}$ ) was added. After 1 h incubation, the wells were rinsed again and sequentially treated with 100  $\mu\text{L}$  of horse radish peroxidase solution and 3,3',5,5'-tetramethylbenzidine (TMB) substrate for 30 min in the dark with a washing step in between. Color formation was observed, and the reaction was stopped by adding 100  $\mu\text{L}$  2 N  $\text{H}_2\text{SO}_4$ . Absorbance was measured at 450 nm using a multi-well plate spectrophotometer. The concentration of albumin synthesized by the cells was calculated from the standard curve and presented as the mean  $\pm$  standard deviation.

The ammonia detoxification ability of HepG2 cells was analyzed by estimating urea in the spent medium using a urea assay kit (Biochain, USA). Fifty microliters of medium, urea standard, and control medium (blank) was added in triplicate to a 96 well plate. The working reagent (200  $\mu\text{L}$ ) was added to each well and incubated for 50 min. The color developed was quantified from the absorbance values at 450 nm, and the concentration of urea was calculated according to the manufacturer's instructions in product sheet.

### 2.7. Characterization of hydrogel

#### 2.7.1. Scanning electron microscopy

Hydrogels were prepared from GelMA and GelMA-FRS(3.4) solutions and lyophilized to obtain discs with dimensions of  $\Phi$  8 mm  $\times$  2 mm. The discs were sputter-coated with gold and imaged using a scanning electron microscope (FEI Quanta 200, Hitachi) to observe the morphology.

#### 2.7.2. Hydrogel porosity and pore size distribution

The porosity of hydrogel was determined using a micro-computed tomography (micro-CT) scanner (Scano Medical, Model#  $\mu\text{CT}$  40, Switzerland). The GelMA and GelMA-FRS(3.4) hydrogel samples having dimensions (9 mm diameter and 10 mm height) was lyophilized and scanned using a cone-beam X-ray source having 5  $\mu\text{m}$  spot size set at 70 kV and 160  $\mu\text{A}$ . 3D reconstruction of the 2D images were done using the software  $\mu\text{CT}$  V6.1. Morphometric analysis was also carried out to determine pores size, percentage pore volume, polymer volume (PV) and total volume (TV). The pore size distribution was plotted using percentage pore volume against pore size. The percentage porosity of the hydrogel was calculated from the equation given below.

$$\text{Porosity (\%)} = \left( \frac{\text{PV}}{\text{TV}} \right) \times 100$$

#### 2.7.3. Compressive modulus of hydrogel

The compressive modulus of hydrogel was determined by compression test using a universal testing machine (UTM) (Bioplus 3345, Instron, USA). Two different sets of cylindrical hydrogels (diameter = 9 mm, height = 10 mm) consisting of 10% GelMA and 10% GelMA-FRS (3.4) were cast in a cylindrical mold by UV-crosslinking. The compression test was performed at  $25^{\circ}\text{C}$  using a 100 N load cell. The samples were compressed at a strain rate of 10 mm/min until a strain of 50% was achieved. The compressive modulus was calculated from the stress-

strain curve.

### 2.8. Formulation of bioink and rheological analysis

The bioink formulation was established for 3D bioprinting of GelMA systems. An ideal polymer system used for 3D bioprinting should exhibit reversible physical gelling and shear-thinning properties. Bioink formulations were obtained using a mixture of GelMA, gelatin, glycerol, and Irgacure 2959 and FRS were obtained. The GelMA-FRS(3.4) formulation consisted of 10% GelMA, 3% gelatin, 10% glycerol, 1% Irgacure 2959 and 3.4 mM ascorbic acid and were subjected to rheological analysis on a Physica MCR 301 Rheometer (Anton Paar, Germany) using cone and plate geometry ( $1^{\circ}$  cone plate, 0.4 mm measuring distance). Viscosity variation of the bioinks was measured with a variable shear rate (0.02–1000 per second) using a light source Omnicure 1000 (Lumen Dynamix LDGI) with a wavelength of 365 nm was routed through a 5 mm optical fiber to illuminate the sample through a quartz crystal stage. This formulation was compared with the basic GelMA-FRS (3.4) polymer solution (10% GelMA, 1% Irgacure 2959 and 3.4 mM ascorbic acid) under similar conditions.

### 2.9. Three-dimensional bioprinting of liver construct

Three-dimensional bioprinting was carried out on a 3D bioprinter (3D Discovery, RegenHU, Switzerland) using GelMA formulation (10% GelMA, 3% gelatin, 10% glycerol, and 1% Irgacure 2959) and GelMA-FRS(3.4) formulations. To prepare the bioinks, approximately  $3 \times 10^6$  HepG2 cells were mixed with 1 mL of the respective GelMA systems, equilibrated at  $20^{\circ}\text{C}$ , and loaded into the extrusion print head of the 3D bioprinter. A tapered plastic nozzle with a 410  $\mu\text{m}$  orifice was also connected. A 3D model square grid pattern (final size of  $10 \times 10 \times 5$  mm) with straight lines and cross-overs was designed using BioCAD software (regenHU, Switzerland), and the construct was bioprinted at a printing speed of 10 mm/s at  $20^{\circ}\text{C}$  into a non-cell-adherent culture plate. The bioprinted constructs were irradiated with 365 nm UV light for 90s. Sufficient culture medium was added to the bioprinted constructs and maintained in a  $\text{CO}_2$  incubator at  $37^{\circ}\text{C}$  for 7 days.

### 2.10. Functional evaluation of 3D bioprinted construct

The photoprotective nature of GelMA-FRS(3.4) formulation on liver-specific functions was determined by analyzing cell viability, albumin, and urea synthesis in comparison to the construct bioprinted using GelMA formulation.

#### 2.10.1. Cell viability of 3D bioprinted liver construct

The proliferation of HepG2 cells in the 3D bioprinted constructs (fabricated using GelMA and GelMA-FRS (3.4) formulations) was compared. The bioprinted constructs were cultured for 24 h and cell proliferation was analyzed by CCK-8 assay on days 1, 3, 5, and 7. The culture medium was replaced with CCK-8 medium (1:10 dilution of CCK-8 reagent in serum-free MEM) and incubated for 2 h. The medium was then transferred to a 96 multi well plate and the absorbance was measured at 450 nm using a multi-well plate reader. The same constructs were further maintained to repeat the CCK-8 assay on the following days.

#### 2.10.2. Functional evaluation of 3D bioprinted liver construct

The functional evaluation of the 3D bioprinted liver construct was confirmed from the amounts of albumin and urea secreted from the construct into the culture medium. Culture medium was collected at the end of 1, 3, 5, and 7 days and stored at  $-80^{\circ}\text{C}$  until used for estimation. The albumin and urea secreted from the 3D bioprinted construct were quantified using the Human Albumin ELISA Kit and Urea Assay Kit, respectively, as described previously.

### 2.11. Statistical analysis

All experiments were performed in triplicates ( $n = 3$ ). The results are presented as mean  $\pm$  standard deviation, and significance was calculated by Student's *t*-test. Statistical significance was set at ' $p$ '  $\leq 0.05$ .

## 3. Results and discussion

### 3.1. Synthesis and characterization of gelatin methacrylamide hydrogel

Gelatin methacrylamide is a biomaterial that is widely used in extrusion bioprinting. Owing to its excellent flow characteristics, biocompatibility, and tunable biodegradability, GelMA has been used for the fabrication of cardiac, skeletal, neuronal, bone, cartilage, and skin tissues [21–27]. We synthesized GelMA from Gelatin Type A based on the method reported by Lee et al., which exclusively converts gelatin to gelatin methacrylamide upon reaction in carbonate – bicarbonate buffer at pH 10.5 [19]. Type A gelatin was selected due to its higher storage modulus over other grades of Gelatin at similar concentration. The carbonate – bicarbonate buffer used in the synthesis ensured that the pH was above the isoelectric point of Type A gelatin aided in neutral free amino group of lysine. The neutrally charged amino groups effectively reacted with MAA, thereby providing a higher degree of functionalization. The presence of two peaks at 5.5 and 5.7 ppm in the  $^1\text{H}$ NMR spectra confirmed the presence of methacrylamide moieties onto gelatin (Fig. 2). The formation of vinylic protons of the methacrylamide moiety confirmed that gelatin was successfully functionalized with gelatin methacrylamide [28]. These acrylamide bonds participate in the photopolymerization of GelMA for hydrogel formation.

The functionalization of gelatin was determined by estimating the unmodified free amino groups in GelMA using a previously reported protocol [20]. A standard curve of gelatin was generated, and the percentage of free amino groups was extrapolated from it (Fig. 3a). The degree of functionalization was calculated from three different batches of GelMA and was found to be approximately  $90 \pm 4\%$  (Fig. 3b).

Morphological analysis of lyophilized GelMA and GelMA-FRS(3.4) was investigated by SEM. All the samples displayed porosity. The pore architecture, however, varied with the addition of FRS and post-

crosslinking (Fig. 4). SEM images also indicated that the porosity of GelMA was uniform in size and shape compared to the crosslinked GelMA and GelMA-FRS(3.4) counterparts (Fig. 4a). Furthermore, scaffolds fabricated from GelMA hydrogels were circular in shape. Scaffolds from crosslinked GelMA and GelMA-FRS(3.4) on the other hand, were elliptical in shape. Nevertheless, the scaffolds were sufficiently porous to accommodate the cells and enable the proper transfer of nutrients and wastes across the scaffold.

The lyophilized GelMA and GelMA-FRS(3.4) hydrogels exhibited porosity of 74.1% and 74.8%, respectively. Fig. 5a shows lyophilized hydrogels used for  $\mu\text{CT}$  analysis. The maximum pore volume of the hydrogels was occupied by pores in the size range of 50–200  $\mu\text{m}$  (Fig. 5b). A similar trend in pore size distribution was observed in both the hydrogel samples.

The compressive moduli of GelMA and GelMA-FRS(3.4) is shown in Fig. 6. The compressive modulus of GelMA and GelMA-FRS(3.4) was  $1.37 \pm 0.24$  kPa and  $0.29 \pm 0.03$  kPa. The addition of FRS lead to 78% reduction in the modulus values of the gel. Nevertheless, both the hydrogels were self-standing and exhibited sufficient handling stability. Hydrogels with lower modulus are known to support better cell viability, adhesion and proliferation [29].

### 3.2. Selection of free radical scavenger concentration

GelMA is a well-studied bioink material for 3D bioprinting with the capacity to form hydrogels when subjected to photo-initiated (mostly through UV irradiation) polymerization. When bioprinted GelMA constructs are irradiated with UV, a large number of free radicals are generated, initiating chain growth polymerization and subsequent cross-linking of GelMA. In the presence of a free-radical scavenger, the extra free radicals generated are scavenged, which might take a toll on the cross-linking of GelMA. Hence, it is imperative to strike a balance between cross-linking efficiency and free radical scavenger concentrations. Furthermore, most photocurable bioinks are based on crosslinking hydrogels with UVA (320–400 nm) and UVB (280–320 nm) radiation. Both A and B versions of UV radiation elicit a physiological effect through the formation of reactive oxygen species inside the cells [30]. In particular, UVA activates endogenous photosensitizers, resulting in the formation of ROS [31]. Moreover, UV exposure impairs the antioxidant

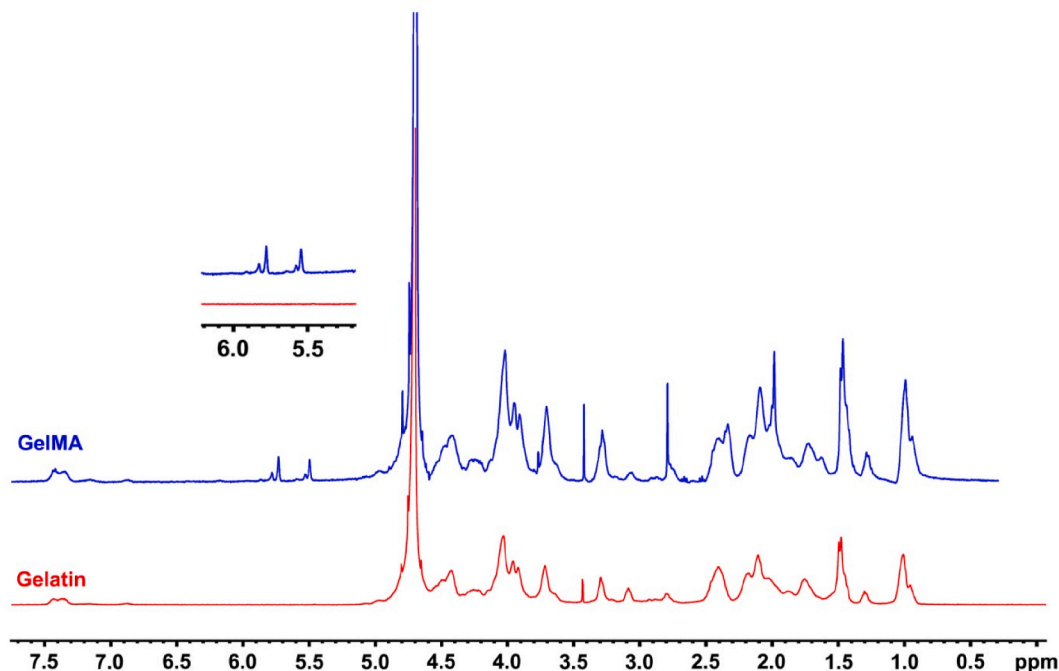
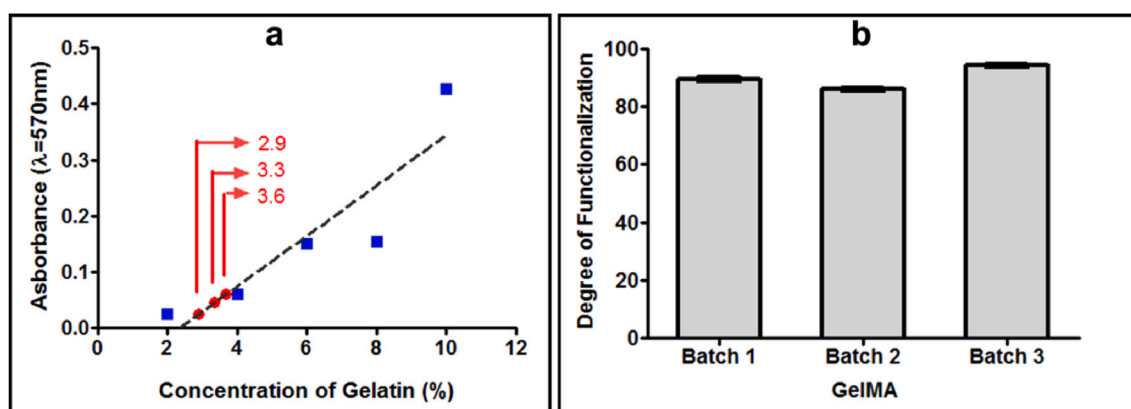
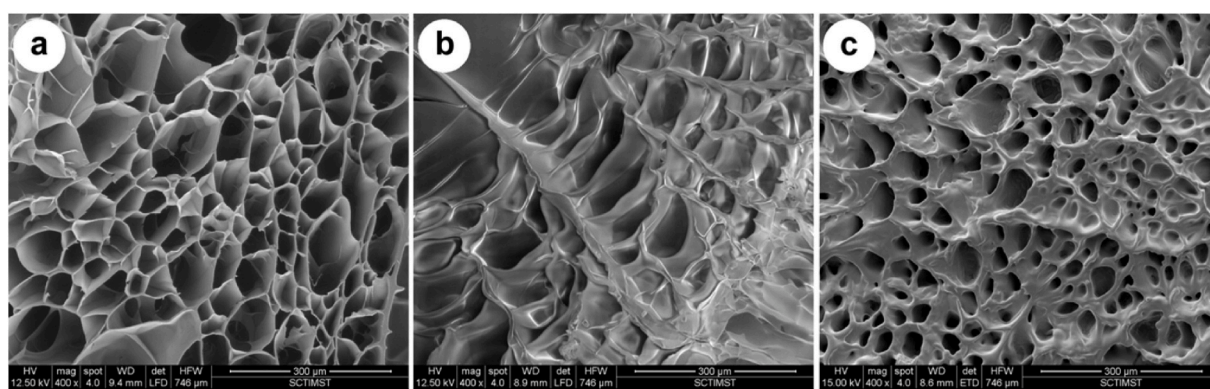


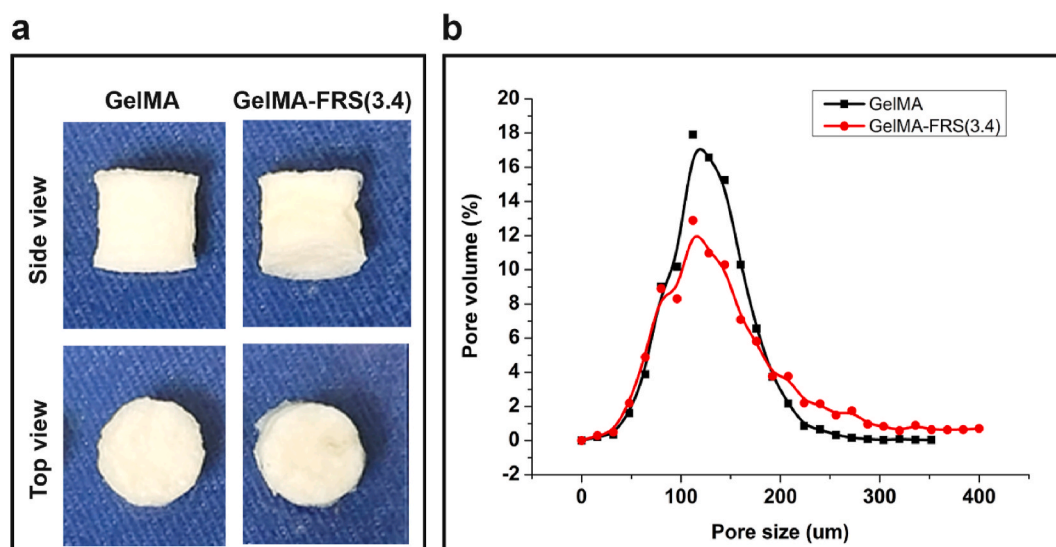
Fig. 2.  $^1\text{H}$ NMR spectra of Gelatin and GelMA. The peaks (5.5 and 5.7 ppm) corresponding to grafted methacrylamide moieties have been zoomed in.



**Fig. 3.** Ninhydrin a) Free amino groups in GelMA extrapolated from free amino groups of gelatin; b) DoF calculated for three different batches of GelMA showing modification efficiency of  $89 \pm 0.6$ ,  $86 \pm 0.5$  and  $94 \pm 0.5\%$ .



**Fig. 4.** SEM images showing porosity of GelMA. (a) GelMA before crosslinking, (b) GelMA-FRS(3.4) before crosslinking and (c) GelMA-FRS(3.4) after crosslinking.



**Fig. 5.** Pore size analysis: (a) Images of the lyophilized GelMA and GelMA-FRS(3.4) hydrogels. (b) Pore size distribution graph plotted using  $\mu\text{CT}$  derived data.

defense mechanisms of the cells. To circumvent the harmful effects of UV, cells are treated with ascorbic acid when exposed to UVA to prevent ROS, GSH, and lipid peroxidation levels [32]. Hence, ascorbic acid was introduced into GelMA as a free radical scavenging system (FRS) to protect cells from free radicals formed during the UV cross-linking of GelMA. This concept of photoprotective effect was studied using a sandwich culture system, as illustrated in Fig. 1. The effect of ascorbic

acid on FRS was monitored by evaluating cell viability, cell proliferation, cell activity, and ROS activity.

### 3.2.1. Cell viability

Fig. 7 shows the results of the viability assay carried out on HepG2 cells entrapped in GelMA-FRS by sandwich culture with various concentrations of FRS. It is clear from the figure that the number of viable

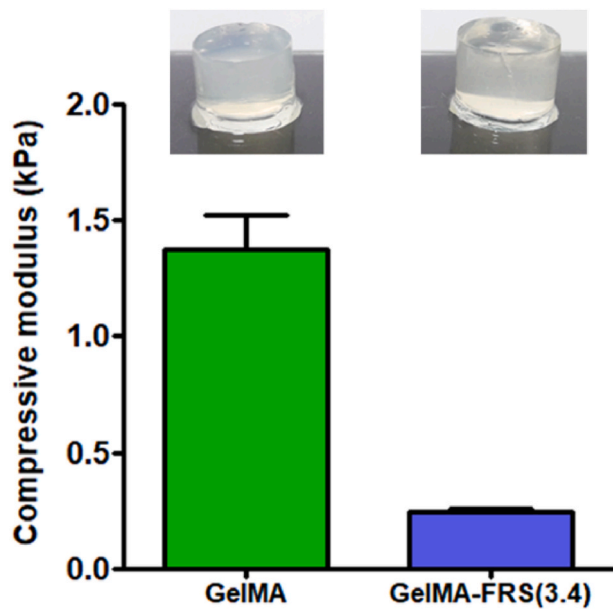


Fig. 6. Compressive modulus of GelMA and GelMA-FRS(3.4) system.

HepG2 cells in the GelMA-FRS sandwich increased with increasing FRS concentration (Fig. 7). The increase in viable cell numbers with increasing FRS concentration is encouraging, as it is a direct testimony of the photoprotective effect of FRS. The photo-protective effect of FRS observed in the cells, however, did not affect the physical cross-linking efficiency of GelMA. Furthermore, GelMA sandwich model the layers adhered to each other without any sign of delamination.

Cell viability assessed by live-dead staining showed an increase in viable cells with increasing concentration of FRS, displaying maximum viability at 3.4 mM. This observation is in line with the report of Tokuda et al., where ascorbic acid had an effect on UV-exposed retina [33]. They found a significant reduction in lactate dehydrogenase release in the presence of ascorbic acid between 1 mM and 3 mM, which confirmed the photoprotective effect.

### 3.2.2. Cell proliferation

The proliferation of HepG2 cells sandwiched between GelMA and GelMA-FRS(X) was quantified using the CCK-8 assay. Cells embedded in GelMA-FRS(1), GelMA-FRS(1.5), and GelMA-FRS(2.2) did not show any increase in proliferation over a period of 7 days (Fig. 8). However, the HepG2 cells sandwiched in GelMA-FRS(3.4), where 3.4 mM of FRS was employed, showed a linear increase in cell number with time. Thus, it was concluded that the minimum concentration of FRS required in the bioink for photoprotective effect is 3.4 mM, without compromising the crosslinking efficiency.

### 3.2.3. Cell activity

The effect of FRS on the functional ability of hepatocytes, such as albumin and urea synthesis, in the GelMA-FRS hydrogel was also evaluated over a period of 7 days. The variations in albumin and urea synthesis at various concentrations of FRS at different time points are shown in Fig. 9. The albumin secretion by cells embedded in GelMA was  $60 \pm 2$  ng/ml (Day 1) and increased linearly to  $86 \pm 4$  ng/ml at the end of 7 days (Fig. 9a). A similar increase in albumin secretion was observed in the cells sandwiched in GelMA with 1, 1.5 mM and 2.2 mM FRS concentrations up to day 5. The albumin secretion in cells sandwiched in GelMA with 2.2 mM FRS was  $111 \pm 7$  ng/ml on day 7. The cells embedded in the gel with FRS 3.4 mM showed albumin synthesis of  $94 \pm 4$ ,  $100 \pm 5$ ,  $131 \pm 16$ , and  $172 \pm 11$  ng/ml on days 1, 3, 5, and 7, respectively. This was very similar to albumin synthesis by the control cells on days 1, 3, and 5. However, the albumin synthesis of control cells on 7th day was  $212 \pm 18$  ng/ml, which was significantly higher than that of the cells in GelMA/FRS 3.4 mM.

The detoxification ability of hepatocytes sandwiched within GelMA and GelMA-FRS hydrogels was analyzed by urea synthesis (Fig. 9b). The average urea synthesis of cells embedded in GelMA with 0, 1, 1.5 mM and 2.2 mM FRS was similar and did not show any significant variation. However, the cells embedded in GelMA with 3.4 mM FRS showed a significant increase in urea synthesis and had values comparable to those of the control cells. This clearly indicates that when GelMA was loaded with 3.4 mM FRS, the functional ability of hepatocytes was improved and was comparable to that of control cells even after entrapping in GelMA and UV crosslinking.

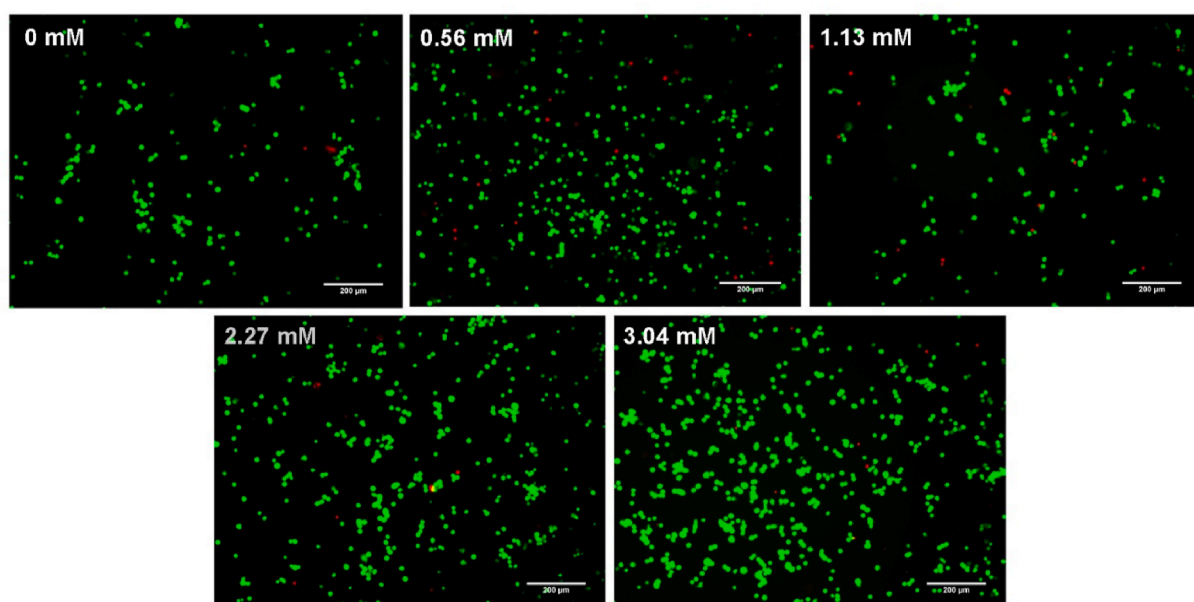
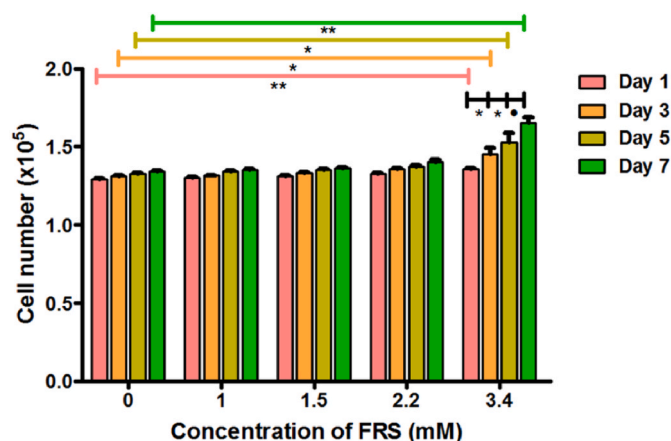


Fig. 7. Viability of HepG2 cells in GelMA with various concentrations of FRS, analyzed by sandwich culture method. Values shown in the figure shows the concentration of FRS present in the base 10% GelMA. The number of viable cells (green) increases with increase in concentration of FRS. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)



**Fig. 8.** Cell proliferation analyzed by CCK-8 assay. GelMA alone (0 mM) and FRS at concentrations 1 mM, 1.5 mM, 2.2 mM and 3.4 mM showed a steady increase in cell number. The values are normalized to cell number of cells in 2D monolayer of the respective days.

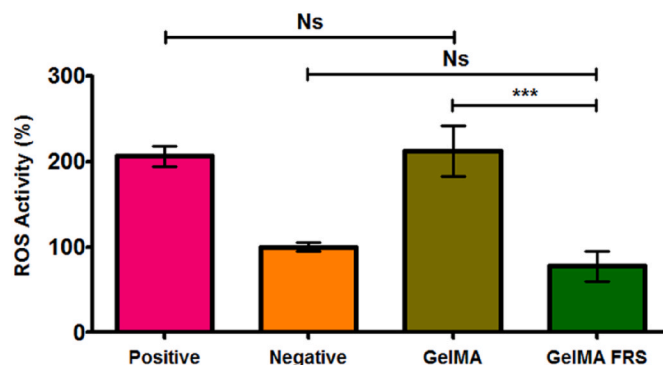
• = ( $P > 0.05$ ), \* = ( $P \leq 0.05$ ), \*\* = ( $P \leq 0.01$ ).

### 3.2.4. Radical oxygen species activity

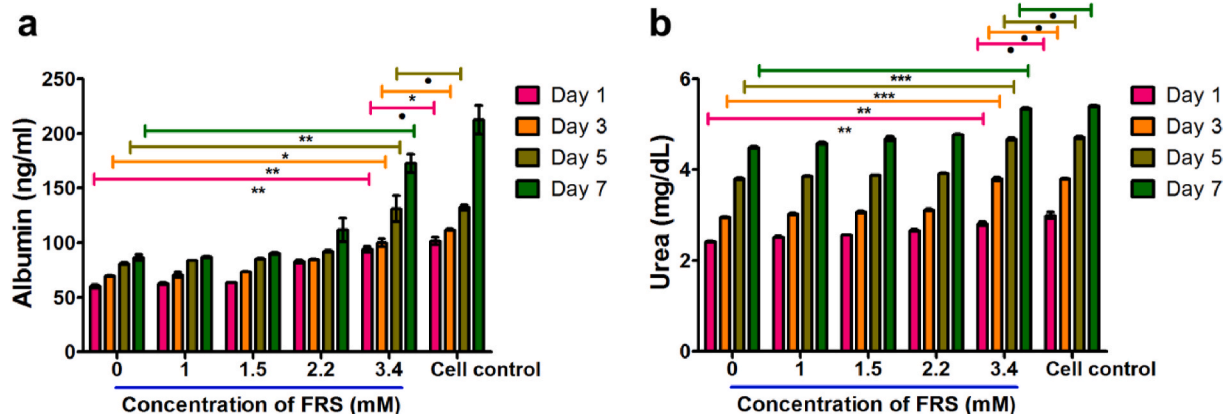
The intracellular ROS activity of HepG2 cells embedded in the GelMA/FRS system (following UV irradiation) was compared with that of the bare GelMA hydrogel (following UV irradiation), cell monolayer exposed to UV (positive control), and cell monolayer unexposed to UV (negative control) (Fig. 10). It can be seen that while ROS activity of cells embedded in GelMA/FRS system was comparable to that of negative control, the ROS activity associated with cells embedded in GelMA was closer to positive control. The ROS values associated with the negative control (cells that were not exposed to UV) represent the normal amount of ROS produced inside the cells. It was encouraging to see that cells embedded in GelMA/FRS ( $77 \pm 42\%$ ) showed values closer to this. On the other hand, the cells embedded in bare GelMA showed ROS activity of  $212 \pm 30\%$ , which was closer to that of the positive control ( $200 \pm 28\%$ ). It is hence clear from these values that the GelMA/FRS system induced protection against the radicals generated during photopolymerization. This is an interesting finding and is particularly relevant when clinically relevant/larger structures are bioprinted and stabilized using photocrosslinking.

The wavelength of light used for crosslinking was selected based on the polymeric system or the photoinitiator added to the formulation. Ultraviolet radiation is classified as UVA (320–400 nm), UVB (290–320 nm), and UVC (200–290 nm), of which UVB and UVC are fatal. One of

the efficient ways to minimize UV-induced damage is by reducing the duration of UV exposure. The wavelengths widely proposed to be safe in 3D bioprinting with respect to cell viability is 365 nm (Irgacure 2959) and 405 nm (LAP or Vis + Ru/SPS) [34,35]. In order to bioprint the clinically relevant size, the duration of crosslinking may be increased. Prolonged exposure of cells to UV causes lesions in genetic content indirectly through the generation of ROS or directly by damaging membrane lipids [36,37]. This limitation can be overcome by nullifying the UV radiation-induced damage using photo-protective methods. Photo-protection generally refers to a biochemical mechanism that helps living systems to prevent/control damage caused by light of harmful wavelengths. Cells neutralize the ROS formed during UV exposure, and any disturbance in ROS production and neutralization will create oxidative stress. Humans and other animals have also developed photoprotective mechanisms to prevent UV photo-damage to the skin, prevent DNA damage, and minimize the downstream effects of oxidative stress [38]. Irgacure 2959 has been shown to be the most suitable UV PI with different sensitivities to different cell types at the same concentration [12,39]. The concentration of PI must be accurately optimized to achieve low cytotoxicity and better crosslinking. It is known that an increase in the concentration of PI results in an increase in cell death due to the free radicals formed during UV irradiation. Irgacure 2959 dissociates into free radicals and initiates the polymerization of functionalized GelMA. The free radical effect of PIs on cells can be counteracted by antioxidants, such as ascorbic acid [40]. Their study used poly (N-isopropylacrylamide) nanoparticles and poly(ethylene glycol) diacrylate crosslinked in the presence of Irgacure 2959 and ascorbic acid,



**Fig. 10.** The comparison of intracellular ROS activity of HepG2 cells due to photocrosslinking of GelMA and GelMA-FRS bioink to normal cells (Negative) and UV exposed (Positive) cells. [Ns = Non significant, \*\*\* =  $p \leq 0.0001$ ].



**Fig. 9.** Functionality assessment of HepG2 cells sandwiched in GelMA and GelMA FRS hydrogels. (a) Albumin secretion by cells sandwiched in GelMA and various concentration of FRS containing GelMA showing 3.4 mM FRS as a formulation that is similar to normal culture system. (b) Urea synthesis indicating the detoxification function did not show any dose related response. GelMA with and without FRS express same urea secretion. • = ( $P > 0.05$ ), \* = ( $P \leq 0.05$ ), \*\* = ( $P \leq 0.05$ ), \*\*\* = ( $P \leq 0.005$ ).

which showed increased cell viability in the presence of antioxidants. The effect of ROS due to the photoinitiator can be best described by measuring intracellular ROS. To the best of our knowledge, this is the first report to explain the intracellular ROS activity of GelMA-based bioink. The aforementioned analysis aided in the effect of FRS concentration on cell viability and liver-specific functions of hepatocytes sandwiched in GelMA. GelMA-FRS(3.4) showed better viability and cell proliferation than GelMA and other GelMA/FRS formulations used in the present study. Hence GelMA/FRS with 3.4 mM FRS, hereafter referred to as GelMA/FRS (3.4), was used in the bioink formulation for all further studies. It has been reported that FRS at concentrations of 4 mM and 10 mM is not cytotoxic and exhibited hepatoprotective effect in cultured rat hepatocytes [41,42].

### 3.3. Formulation of bioink and rheological analysis

The base material in the bioink formulation was 10% GelMA in serum-free culture medium. To enhance printability, 3% gelatin and 10% glycerol were added as reported by Kang et al. [43]. While gelatin imparts thermosetting properties for the bioink, which enables one to carry out printing at temperatures ranging from 18 to 21 °C, glycerol functions as a plasticizer and helps to reduce the friction of the bioink at the nozzle and aids the bioink to drop on the printing platform with ease. Both gelatin and glycerol were gradually washed out during the post-printing culture [44].

Systematic analyses of the rheological properties were carried out to understand the gelation kinetics of the bioink solutions. The effects of time, temperature, shear stress, and shear rate on the gelation process of GelMA solutions were evaluated. The gelation temperature of the polymer solutions was assessed by cooling the GelMA solutions from 37 °C to 2 °C at a cooling rate of 5 °C/min (Fig. 11). It was noticed that as the temperature decreased, the viscosity increased gradually for both GelMA-FRS(3.4) and GelMA-FRS(3.4) formulations. At the gelation temperature (20–30 °C), there was a rapid increase in viscosity. The viscosity values showed nearly a 100 fold increase from 0.0156 Pa s (24 °C) to 9.991 Pa s (17 °C) for GelMA-FRS(3.4), and 0.00283 (24 °C) and 4.0155 (15 °C) for the GelMA-FRS(3.4) solution, after which the curve plateaued, demonstrating the stability of the gels formed. The precise gelling temperature of the GelMA solutions was determined from the derivative of the cooling curves. While GelMA-FRS(3.4) solutions gelled at 18.6 °C, the gelling temperature of GelMA-FRS(3.4) formulation was 20.5 °C. This is expected because of the presence of native

gelatin in the GelMA-FRS(3.4) formulation that aids not only in increasing the viscosity but also in improving the thermo-responsive property of the bioink.

In rheology, shear-thinning refers to the non-Newtonian behavior of fluids, where viscosity decreases under shear [45]. This property is particularly relevant to bioinks, as it determines continuous extrudability, high shape fidelity, and mechanical resilience [46]. Shear thinning of bioinks prevents clogging at nozzles during the printing process and aids in the reduction of shear stress that can lead to cell damage [47]. Furthermore, to attain constructs with a better resolution and higher aspect ratio, the printed scaffold should be mechanically sturdy and should not collapse on its own weight. Instant gelation after deposition or partially pre-gelled hydrogels is also preferred to ensure high structural integrity [48].

The shear thinning behavior of the polymer solutions is evident from the viscosity variations exhibited at different shear rates (Fig. 12). Both GelMA-FRS(3.4) and GelMA-FRS(3.4) formulations exhibited a decrease in viscosity with increasing shear rates, with GelMA-FRS(3.4) formulation showing a higher tendency of shear thinning. At a given shear rate, the GelMA-FRS(3.4) formulation displayed a higher viscosity than the GelMA-FRS(3.4) solution.

The gelation characteristics of the GelMA-FRS(3.4) bioink solution and formulation were also investigated in the presence of UV radiation. Fig. 13 shows the variation in the complex viscosity and storage modulus of the bioink with time during the crosslinking process. It is clear from the figure that both the viscosity and storage modulus increase with time. The increase was drastic at the beginning, indicating that the polymer solution underwent cross-linking. As the cross-linking moieties were consumed, the viscosity did not increase. A similar trend was shown by the storage modulus data, where the modulus increased and reached a plateau. With time, the elastic modulus finally dominates, and the solutions become more elastic than viscous. This also indicates that the gel retains sufficient elastic recoil, which is important for maintaining the shape of the gel after extrusion to avoid variations in product size.

### 3.4. Three-dimensional bioprinting of liver construct

Three-dimensional bioprinting of complex tissues, such as the liver, has been an interesting field since the time when this technology started popular. The main requirement for 3D bioprinting of the liver construct is the maintenance of hepatocyte-specific functions. Various bioinks have been proposed for bioprinting the liver, such as collagen-chitosan, alginate, and decellularized extracellular matrix [49–51]. Here, we report gelatin methacrylamide as the bulk polymer of the bioink with a photoprotective FRS additive to improve and maintain viability, proliferation, and liver-specific functions. The gelatin used for this study provided a better resolution and shape fidelity at room temperature and during 3D bioprinting. The gist of tissue engineering of the liver construct is the functional capability of metabolic activity. The design used for bioprinting is suitable for efficient mass transfer. The grid pattern enabled the diffusion of metabolites effectively, which could be measured in a static culture medium. The bioprinted construct did not show any swelling property, which is a unique characteristic expected in design-based biofabrication.

The bioink formulation was selected based on the outcome of the printed construct using the CAD design. The bioink was extruded with very smooth filaments with a width of 410 µm. The final printed construct was exactly the same size as the input design and was structurally stable before and after UV curing (Fig. 14).

### 3.5. Functional evaluation of 3D bioprinted construct

The metabolic activity of HepG2 cells in constructs 3D bioprinted with GelMA and GelMA-FRS(3.4) formulation was evaluated at two time points (day 1 and day 7) using the CCK-8 assay. The metabolic activity of

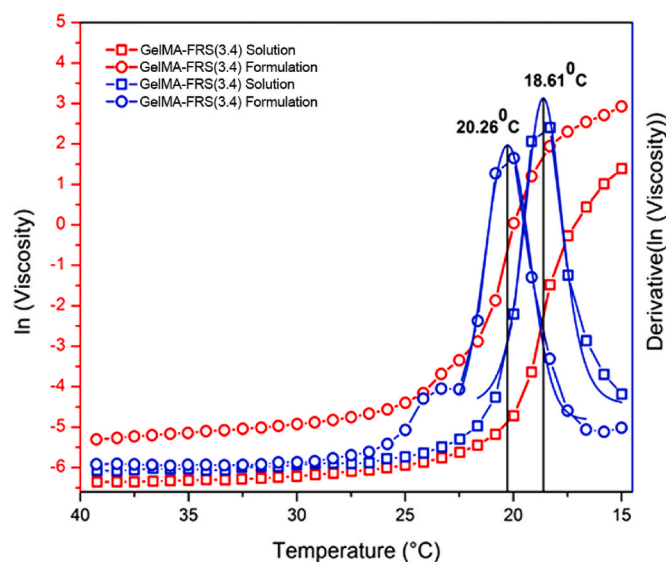


Fig. 11. Dependence of viscosity on temperature of GelMA-FRS(3.4) solution and GelMA-FRS(3.4) formulation.

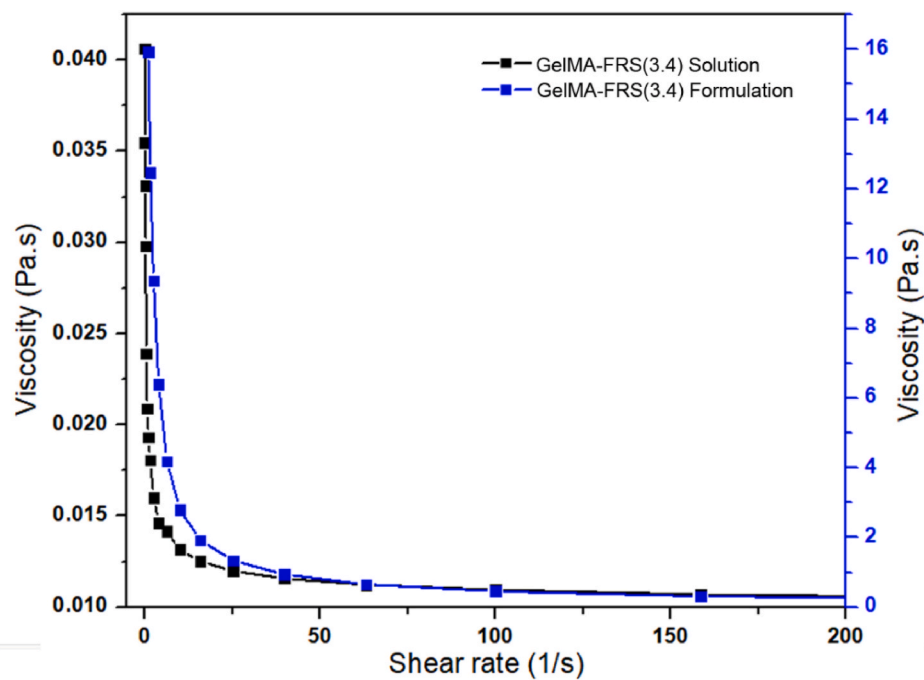


Fig. 12. Dependence of viscosity on shear rate of GelMA-FRS(3.4) solution and GelMA-FRS(3.4) formulation.

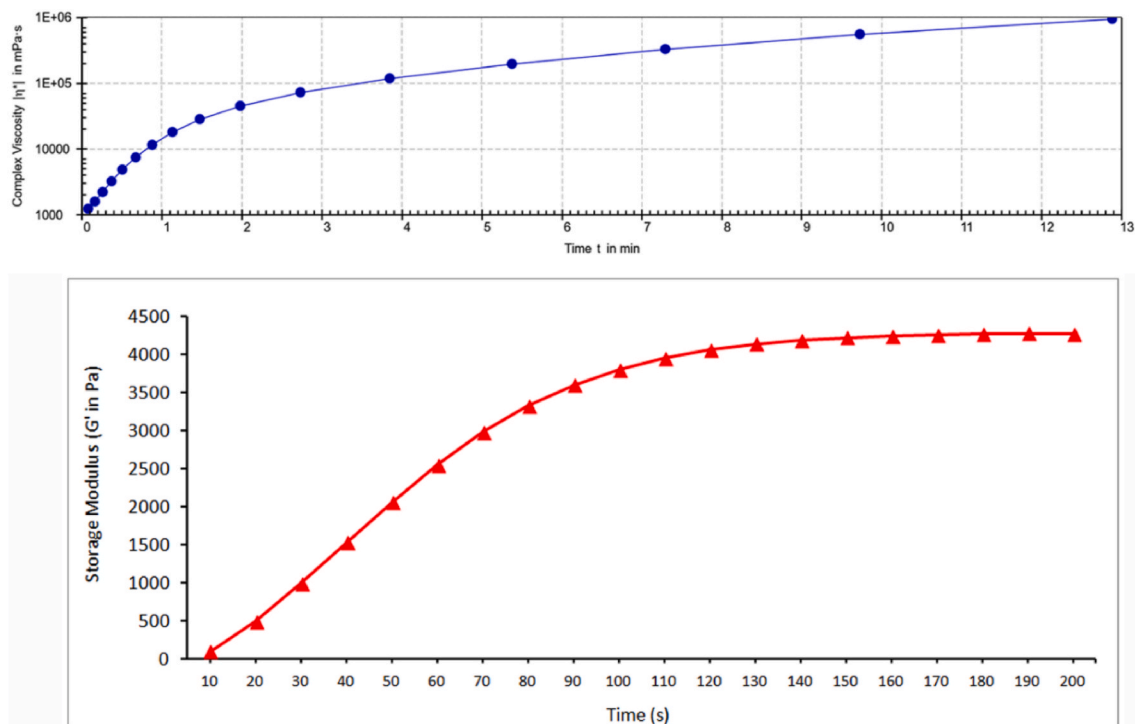
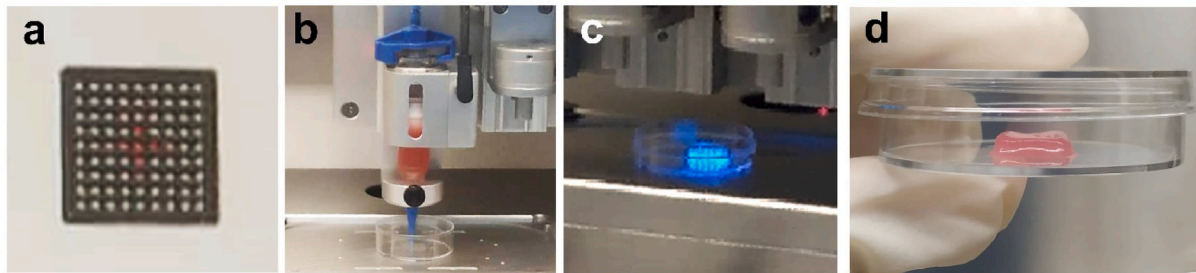


Fig. 13. Variation of complex viscosity and storage modulus of GelMA-FRS(3.4) formulation with time.

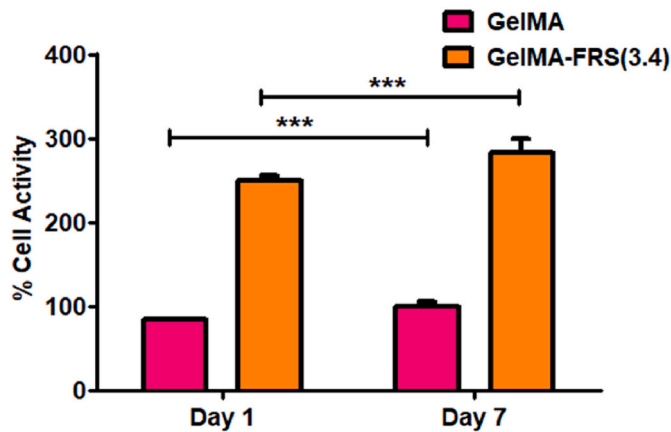
HepG2 cells in GelMA-FRS(3.4) bioink was considerably higher than that of HepG2 cells in GelMA (Fig. 15). At the end of 7 days, cell activity in the 3.4 mM FRS bioink formulation was significantly higher than that printed using GelMA alone. This clearly shows that the presence of FRS in the bioink maintained and improved the cell activity after cross-linking with UV.

The liver-specific functions expressed by the 3D bioprinted constructs showed that cells printed in FRS containing GelMA were more

functionally capable than those printed using GelMA alone (Fig. 16). Albumin synthesis in the construct bioprinted with GelMA FRS was more than that in constructs printed in GelMA bioink alone (Fig. 16). The protein synthesis on day 1 in GelMA-FRS(3.4) was  $50 \pm 2$  ng/ml and increased up to  $128 \pm 3$  ng/ml on 7th day in a linear fashion. A similar trend of increase was observed in the construct printed using GelMA bioink, but a lower albumin synthesis of  $32 \pm 3$  ng/ml and increased up to  $91 \pm 8$  ng/ml on 7th day. The percentage increase in albumin



**Fig. 14.** 3D bioprinting of liver construct using GelMA-FRS(3.4) bioink. (a) The CAD design showing the dimensions of the construct. (b) Customized 3D Bioprinter for extrusion of hydrogels, (c) Image showing 3D bioprinting in action, (d) Bioprinted liver construct.



**Fig. 15.** Cell proliferation of HepG2 cells assessed by CCK-8 assay in 3D bioprinted in GelMA and GelMA-FRS(3.4) bioink formulations. The values are normalized to cell activity of construct in GelMA bioink. [\*\*\* =  $p \leq 0.001$ .]

synthesis relative to day 1 in the GelMA-FRS(3.4) construct was  $27 \pm 17\%$ ,  $86 \pm 18\%$ , and  $154 \pm 5\%$  on days 3, 5, and 7, respectively. On the other hand, the percentage increase in albumin synthesis by cells in GelMA on days 3, 5, and 7 was  $22 \pm 6\%$ ,  $98 \pm 5\%$ , and  $185 \pm 24\%$ , respectively. On the other hand, the urea showed  $12 \pm 2$ ,  $30 \pm 5$ ,  $40 \pm 0$ , and  $33 \pm 3\%$  increase in GelMA-FRS(3.4) compared to GelMA bioink.

Cell proliferation in GelMA-FRS(3.4) bioink was significantly higher than that in GelMA. The functional capability of the construct as a biofabricated liver was evidenced by albumin and urea synthesis. The addition of FRS into GelMA clearly maintained or improved cell viability and functionality. The construct with functional hepatocytes is also a promising candidate for *in vitro* hepatotoxicity testing of drugs and small molecules. It is also useful to model disease-specific tissues for drug testing. Even though crosslinked with UV, the peculiar property of the

bioink in maintaining intracellular activity increases its potential for use in cancer research and preclinical evaluation of drugs.

#### 4. Conclusions

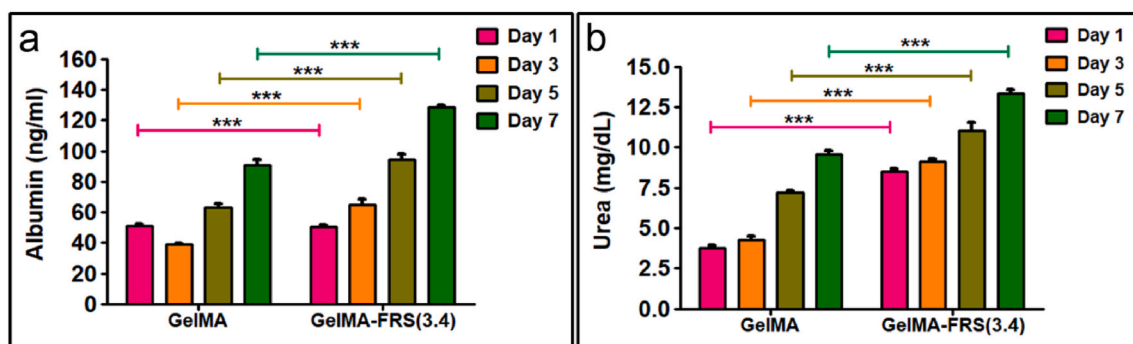
This study reports for the first time the use of a photoprotective GelMA-based bioink for 3D bioprinting of liver constructs. The results showed that radical scavenging during photocrosslinking improved the viability and functionality of the bioprinted constructs. The liver construct biofabricated using the photoprotective GelMA-FRS(3.4) bioink could be a simple and effective approach to enhance liver-specific functions in bioengineered liver tissues. The hepatocytes encapsulated within the novel formulation of bioink by UV irradiation clearly demonstrated the effective nullification of intracellular ROS and maintained liver-specific functions. The cells in the 3D bioprinted construct expressed more than 150 times the activity and improved cell functionality compared to normal GelMA. Our future studies are expected to address the physiological capabilities and structural organization of 3D liver tissues bioprinted with heterogeneous primary cells. We expect that this bioink formulation, when evaluated further based on regulatory standards, will be a realistic approach to develop transplantable liver constructs to overcome the problems of organ shortage in the future.

#### Conflict of interest

None.

#### CRediT authorship contribution statement

**Roopesh R. Pai:** Writing – original draft, Acquisition of data, Formal analysis, Writing – original draft, Critical revision for important intellectual content. **Shilpa Ajit:** Acquisition of data, Formal analysis, Writing – original draft. **Anupama Sekar J:** Acquisition of data, Formal analysis, Writing – original draft. **Sarath S. Nair:** Writing – original



**Fig. 16.** Liver specific function exhibited by the Liver construct 3D bioprinted in GelMA-FRS (3.4 mM) compared with GelMA (0 mM). (a) Albumin synthesis and (b) Urea synthesis was more in GelMA-FRS(3.4) than hepatocyte printed in GelMA. [\*\*\* =  $p \leq 0.001$ .]

draft. **Anil Kumar P.R.:** Writing – original draft, Acquisition of data, Formal analysis, Writing – original draft, Critical revision for important intellectual content, Methodology, Final approval of the version to be published. **Shiny Velayudhan:** Writing – original draft, Acquisition of data, Formal analysis, Writing – original draft, Critical revision for important intellectual content, Resources, Final approval of the version to be published.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.bprint.2022.e00214>.

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